

N+ - Network Fundamentals and building network

IP Addressing & Subnetting

[Q-1] List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Class	IP Address Range	Example Device/Service
A	1.0.0.0 – 126.255.255.255	Large organization or ISP network
B	128.0.0.0 – 191.255.255.255	University or medium-sized company network
C	192.0.0.0 – 223.255.255.255	Home Wi-Fi router or small business network
D	224.0.0.0 – 239.255.255.255	Multicast applications such as video streaming
E	240.0.0.0 – 255.255.255.255	Experimental and research purposes

[Q-2] Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses.

Hint:
Show your working steps for each calculation.

- IP Address = **192.168.10.45**
- Subnet Mask = **255.255.255.0 (/24)**

Step 1: Determine the Network Address

For a **/24** subnet, the first 24 bits represent the network portion and the last 8 bits represent the host portion.

Set all host bits to **0**:

Network Address = 192.168.10.0

Step 2: Determine the Broadcast Address

Set all host bits to **1**:

Broadcast Address = 192.168.10.255

Step 3: Determine the Valid Host Range

- First usable host: **192.168.10.1**
- Last usable host: **192.168.10.254**

Final Answer

Item	Address
Network Address	192.168.10.0
Broadcast Address	192.168.10.255
First Host	192.168.10.1
Last Host	192.168.10.254

[Q-3] Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Given Network: 10.0.0.0/24

To create **4 equal subnets**, divide the /24 network into four **/26 subnets**.

10.0.0.0/24 10.0.0.0/24 10.0.0.0/24

Subnet Allocation

Group	Network Address	Subnet Mask	Usable Hosts
Delivery Partners	10.0.0.0/26	255.255.255.192	62
Restaurants	10.0.0.64/26	255.255.255.192	62
Customers	10.0.0.128/26	255.255.255.192	62
Admins	10.0.0.192/26	255.255.255.192	62

[Q-4] You are given the following scenario: A college WIFI network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.
Constraint: Show your calculation steps and final answers.

❑ **Network Address = 172.16.0.0/16**

❑ **Required Subnets = At least 20**

Step 1: Determine Borrowed Bits

Need at least 20 subnets:

- $2^4 = 16$ subnets (not enough)
- $2^5 = 32$ subnets (enough)

Therefore, borrow **5 bits** from the host portion.

Step 2: Calculate New Subnet Mask

Original prefix = **/16**

Borrowed bits = **5**

New prefix = **/21**

Subnet mask:

255.255.248.0

Step 3: Calculate Hosts per Subnet

Remaining host bits:

$$32 - 21 = 11$$

Hosts per subnet:

$$2^{11} - 2 = 2048 - 2 = 2046$$

So, each subnet can support **2046 usable hosts**.

Final Answer

Item	Value
New Subnet Mask	255.255.248.0 (/21)
Number of Subnets	32
Hosts per Subnet	2046

[Q-5] A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

Your device is not connecting because the IP address **192.168.1.300** is invalid. In an IPv4 address, each number (octet) must be between **0 and 255**. Since **300** is greater than **255**, the address cannot be used on a network.

A valid IP address in the same subnet would look like **192.168.1.x**, where **x** is typically between **1 and 254**. Examples include **192.168.1.10**, **192.168.1.50**, or **192.168.1.200**.